

Task 2 – Feature Engineering, Model Optimization & Performance Comparison

OBJECTIVE

To improve machine learning model performance using feature scaling and compare multiple regression algorithms on the California Housing Dataset.

DATASET USED

California Housing Dataset

TARGET VARIABLE

HousePrice (Median House Value)

FEATURES

- Median Income
- House Age
- Average Rooms
- Average Bedrooms
- Population
- Average Occupancy
- Latitude
- Longitude

METHODOLOGY

1. Loaded the California Housing Dataset.
2. Separated features and target variable.
3. Applied StandardScaler for feature scaling.
4. Performed Train-Test Split.
5. Trained multiple regression models.
6. Evaluated models using RMSE and R^2 Score.

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	HousePrice
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422

Feature Scaling Completed

TRAIN TEST SPLIT

The dataset was divided into training and testing sets to evaluate model performance on unseen data.

Training Data Shape: (16512, 8)
Testing Data Shape: (4128, 8)

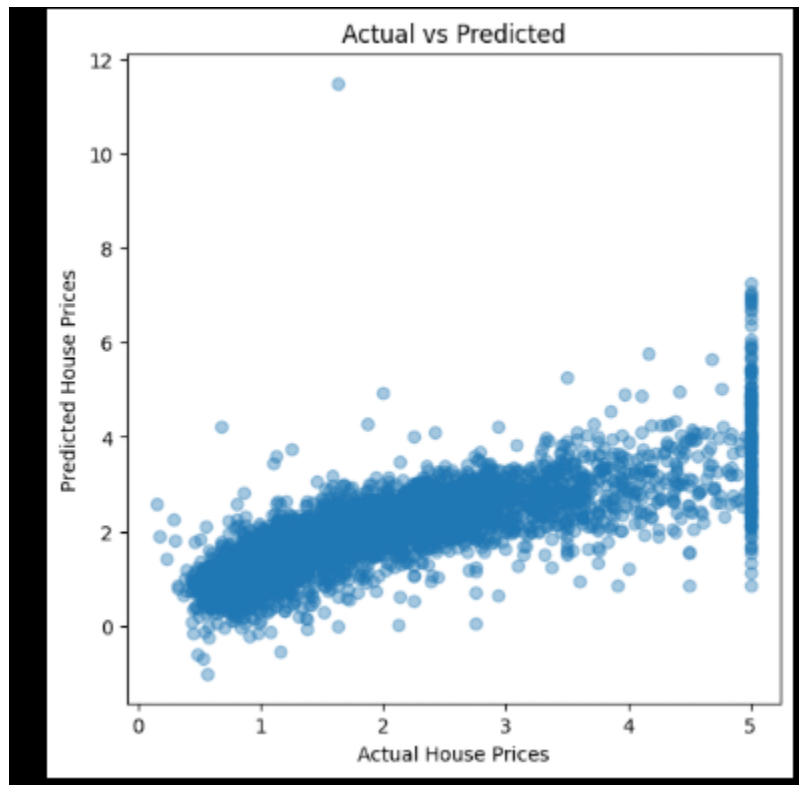
MODEL COMPARISON TABLE

Model	RMSE	R ² Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree	0.724234	0.599732

	RMSE	R2 Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree	0.724234	0.599732

VISUAL PERFORMANCE VALIDATION

Actual vs Predicted values were plotted to visually evaluate model performance.



BEST MODEL

Decision Tree Regressor

REASON

- Lowest RMSE (0.724234)
- Highest R^2 Score (0.599732)

CONCLUSION

Three regression models were trained and evaluated on the California Housing Dataset. The Decision Tree Regressor achieved the lowest RMSE (0.724234) and the highest R^2 Score (0.599732). Therefore, it was selected as the best-performing model among the evaluated algorithms. This task demonstrated the importance of feature scaling, model comparison, and performance evaluation in machine learning workflows.